Problem-Solution Fit

Date	15 July 2026
Team ID	SWTID-2026-3016
Project Name	Credit card approval system
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

1. CUSTOMER SEGMENT(S) [CS] • Individuals applying for a credit card • Banks and financial institutions • Loan/Credit officers	**6. CUSTOMER LIMITATIONS** EG. BUDGET, DEVICES **[CL]** • Limited financial knowledge • Lack of credit history • Time constraints • Internet/device availability	**5. AVAILABLE SOLUTIONS** PROS & CONS **[AS]** **Manual Verification** ✓ Human judgment ✕ Slow and error-prone **Traditional Rule-Based Systems** ✓ Simple to implement ✕ Less accurate and inflexible
2. PROBLEMS / PAINS + ITS FREQUENCY **[PR]** • Manual approval process is slow (High) • Inconsistent decision-making (Medium) • Long waiting time for applicants (High)	**9. PROBLEM ROOT / CAUSE [RC]** • Large number of applications • Time-consuming manual verification • Human bias and inconsistency • Difficulty analyzing multiple applicant factors simultaneously	**7. BEHAVIOR** + ITS INTENSITY **[BE]** • Frequently checks eligibility before applying (High) • Wants quick decisions (High) • Compares different banks (Medium)
3. TRIGGERS TO ACT [TR] • Customer applies for a credit card • Need for instant eligibility check • Bank wants to automate	**10. YOUR SOLUTION [SL]** Develop a **Machine Learning-based Credit Card Approval Prediction System** using **Random**	**8. CHANNELS OF BEHAVIOR [CH]** ONLINE Bank websites, Mobile apps, Credit card portals

approvals	**Forest** and **Flask** to analyze applicant details and predict approval/rejection instantly. The system provides faster, consistent, and data-driven decisions through an easy-to-use web interface.	
4. EMOTIONS BEFORE / AFTER **[EM]** **Before:** Anxious, uncertain, hopeful **After:** Happy if approved; disappointed but informed if rejected		OFFLINE Bank branches, Customer service representatives